BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: Chemistry Paper–IX (02) (22)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** Chemistry Paper–IX (02) (22)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The oxidation state of manganese in $KMnO_4$ is....

(A) +7 (B) +2

(C) +3 (D) +4

Q2. Anion is formed by gaining one or more.....

(A) Neutron (B) Proton

(C) Electron (D) Positron

Q3. The oxidation number of oxygen (O) in H_2O is.....

(A) 1 (B) 2

(C) –2 (D) –1

Q4. Uranium has..... isotopes.

(A) 2 (B) 3

(C) 4 (D) 5

Q5. The example of tri-atomic molecule is.....

(A) CO_2 (B) O_2

(C) CH_4 (D) NH_3

Q6. Oxide of calcium "CaO" is.....

(A) Acidic
(B) Basic
(C) Amphoteric
(D) Neutral

Q7. Fluorine has an electronic configuration, 2,7 and oxygen, 2,6. The formula of fluorine oxide will be

(A) FO
(B) F_2O
(C) FO_2
(D) F_2O_2

Q8. Halogens react with metals to form.....

(A) Halides
(B) Oxide
(C) Halogen sulphides
(D) Hydrogenated compound

Q9. Molarity is the number of moles of solute dissolved in.....

(A) 1 kg of solvent
(B) 1 kg of solution
(C) 1 dm^3 of solvent
(D) 1 dm^3 of solution

Q10. Ink spreads in water because of.....

(A) Vapour pressure
(B) Expansion
(C) Diffusion
(D) Compressibility of water

Q11. Which of the given elements is in the same family as fluorine?

(A) Silicon
(B) Antimony
(C) Iodine
(D) Arsenic

Q12. The maximum number of electrons in third energy level is:

(A) 10
(B) 18
(C) 32
(D) 64

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Answer any EIGHT parts. Each part carries FOUR marks.

Q1. "Like dissolve like", elaborate the statement.

Q2. Why gas is compressible but solid is not compressible? Give reason.

Q3. What is the atomic number of an element? How does it differ from the mass number?

Q4. State the physical properties of metals.

Q5. Calculate the molarity of a solution containing 7.50 mol of $CaCO_3$ in enough water to make 1.50 dm^3 of solution.

Q6. Write down four characteristics of ionic compound.

Q7. How many moles of H_2SO_4 are present in 0.500 dm^3 of 0.150 M of H_2SO_4 solution?

Q8. Why NaOH is strong but NH_4OH is weak electrolyte?

Q9. How does ionization energy values vary in groups?

Q10. Define electroplating and give any three processes of electroplating.

Q11. Why the s-block elements have two groups only?

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

Q12.

(a) [4] What is the main distinction between ionic and covalent bonding? Explain with examples.

(b) [3] Differentiate between atomic radii and covalent radii.

Q13.

(a) [4] Draw the Lewis structure for the given: (i) CCl_4 (ii) BF_3

(b) [3] Write down factors which affect vapour pressure.

Q14.

(a) [3] How would you experimentally verify Boyle's Law?

(b) [4] Define solubility. What is the effect of temperature on solubility?

Q15.

(a) [4] What happens during displacement reaction in halogens?

(b) [3] Differentiate between unsaturated, saturated and super saturated solution.